

DSA (C++) Short Notes

1. Arrays

Contiguous memory storage. Access $O(1)$. Useful for indexing and iteration.

2. Stack (C++ STL)

LIFO structure using stack. Operations: `push()`, `pop()`, `top()`.

3. Queue

FIFO structure using queue. Operations: `push()`, `pop()`, `front()`.

4. Linked List

Dynamic nodes using pointers. Types: Singly, Doubly. Efficient insertion/deletion.

5. Trees

Binary Tree, BST. Traversals: inorder, preorder, postorder. Used in hierarchies.

6. Graphs

Represented using adjacency list/vector. Traversal: BFS (queue), DFS (recursion/stack).

7. Sorting Algorithms

Bubble Sort $O(n^2)$, Merge Sort $O(n \log n)$, Quick Sort $O(n \log n \text{ avg})$.

8. Searching

Linear Search $O(n)$, Binary Search $O(\log n)$ (requires sorted array).

9. Recursion

Function calls itself. Needs base condition. Used in DFS, tree problems.

10. STL in DSA

Important: vector, set, map, unordered_map, priority_queue.

11. Time Complexity

Big-O notation: $O(1)$, $O(n)$, $O(\log n)$, $O(n \log n)$, $O(n^2)$.